

How Far Can 16 GB Go? Characterizing Memory-Efficient LLM Fine-Tuning on Consumer Apple Silicon

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Abstract

We map the fine-tuning feasibility boundary of a 16 GiB Apple M4 Mac mini with a preregistered benchmark: 118 supervised runs over the Qwen3 family (0.6B–14B), BF16 LoRA vs. 4-bit QLoRA, context lengths 512–8192, LoRA ranks 4–32, and micro-batches 1–8, each with full provenance (git commit, model revision, per-second swap sampling, SHA-256 manifests). We find: (1) 4-bit QLoRA cuts peak training memory to $0.54\text{--}0.73\times$ of BF16 and extends TRAINABLE reach to 8B, while step time stays within the preregistered $\pm 25\%$ equivalence margin at all three paired scales; (2) boundaries are joint properties of model and system state — the same 4B BF16 or 14B 4-bit configuration that trains in minutes at zero swap residency stalls at 50% completion or makes zero progress when several GiB of swap are resident; (3) peak memory grows *sublinearly* with model size (log-log slope 0.57 for 4-bit, $R^2=0.99$) because fixed runtime overhead dilutes weight costs, while step time grows linearly (slope 1.00); (4) context length, not model scale, is the binding constraint: 2048-token contexts complete but raise peak memory beyond physical RAM, and 4096-token contexts are killed before their first step; (5) micro-batch scaling fails outright at batch 8 (consistent SIGKILL with system swap driven to ~ 20 GiB) and *reduces* throughput at every trainable size. All numbers are generated programmatically from immutable raw results; seven preregistration deviations are reported in full.

1 Introduction

Consumer Apple Silicon machines ship with a single 16 GiB unified-memory pool that the GPU shares with the OS. Whether, and under what conditions, such a machine can fine-tune modern LLMs is usually answered anecdotally. This paper answers it with a preregistered, failure-inclusive benchmark: every configuration in a frozen matrix was run under an experiment

supervisor that recorded provenance, per-second swap sampling, and complete logs, and every completed or failed run is retained as evidence.

Our contributions: (i) a preregistered feasibility map for Qwen3 0.6B–14B under BF16 LoRA, 4-bit QLoRA, and (exploratory) full fine-tuning; (ii) quantified memory/time trade-offs with frozen equivalence margins; (iii) context-length and micro-batch boundaries expressed as intervals with system-state confounders documented; (iv) an audit showing that TRAINABLE is a joint property of the model and the machine’s memory-residency state, not of the model alone; and (v) a full reproducibility chain from raw immutable results to every number in this paper, including seven logged deviations from preregistration.

2 Background and Related Work

2.1 Apple Silicon and Unified Memory

Apple Silicon SoCs present a single physical memory pool shared by CPU and GPU. `mlx` [9] is an array framework that targets this architecture directly; `mlx-lm` [12] provides LLM fine-tuning on top of it. Prior profiling of Apple Silicon training reports substantial gaps to NVIDIA GPUs and identifies page faults and kernel-launch overhead as system-level contributors [7]. Benchmarking of MLX itself has so far focused on inference latency for encoder models [1]; consumer-device LLM stacks beyond MLX include `llama.cpp` [8].

2.2 MLX

The MLX framework exposes Metal-backed computation with lazy evaluation and a composable optimizer API [9]. Our measurement validation (Sec. 3.6) quantifies why explicit synchronization is required for per-step timing under lazy execution.

2.3 LoRA and QLoRA

LoRA [11] parameterizes weight updates with low-rank adapters; QLoRA [5] backpropagates through 4-bit quantized frozen weights, following the broader line of LLM quantization work [4]. Both reduce the memory footprint relative to full fine-tuning [10, 3], and GaLore [14] targets the same budget from the optimizer-state side. These methods are largely evaluated on data-center GPUs; how they behave inside a 16GB unified-memory budget is the subject of this paper.

2.4 Resource-Constrained LLM Fine-Tuning

Prior systems work has profiled Apple Silicon for ML training end-to-end [7] and benchmarked MLX inference for encoder models [1], but the fine-tuning feasibility boundary on a fixed consumer unified-memory budget — which model scales, precisions, and context lengths admit *completed* training steps, and at what cost — has not, to our knowledge, been mapped by a preregistered benchmark. Closest in spirit, memory-efficiency methods such as GaLore [14] and QLoRA [5] report feasibility on GPU hardware where CPU/GPU memory are separate; unified memory changes the failure mode (page thrumbling rather than allocator exceptions), which motivates our measurement design.

3 Methodology

3.1 Research Questions

We study:

- **RQ1** How do memory, step time, and throughput of BF16 LoRA and 4-bit QLoRA scale with model size?
- **RQ2** How far does 4-bit quantization extend the TRAINABLE boundary?
- **RQ3** Where is the context-length boundary?
- **RQ4** Does the memory saving of quantization come with a measurable training-time penalty?
- **RQ5** How large is the gap between “training can start” and PRACTICAL fine-tuning?

3.2 Hardware and Software

All experiments run on a single Apple Mac mini with an Apple M4 SoC, 16 GiB unified memory, and macOS 26.5; the exact preflight-recorded environment is in Table 1. Software was locked with uv: Python 3.13.11, mlx 0.32.2, mlx-lm 0.31.3. The M4

Table 1: Hardware, software, and model revisions (provenance from raw experiment records).

Hardware	
Chip	Apple M4
Memory	16 GiB unified
CPU cores	10P / 10L
OS	macOS 26.5 (build 25F71)
Python	3.13.11 (uv-managed venv)
MLX / mlx-lm	0.32.2 / 0.31.3
Training data	ultrachat.200k@8049631c, frozen 2048/32 subset
Qwen3 models	
14B-4bit	resolved revision (8-char prefix)
Qwen/0.6B	a4d9b2df...
0.6B-4bit	c1899de2...
Qwen/1.7B	73e3e38d...
1.7B-4bit	70d244cc...
4B-4bit	3b1b1768...
Qwen/4B	4dc3d10...
8B-4bit	1cfa9a72...
	545dc425...

generation context is from Apple’s release material [2].

3.3 Models

We use the Qwen3 family [13]: instruction-tuned chat models (not base models) at 0.6B/1.7B/4B in BF16 and 0.6B–14B as pre-quantized 4-bit mlx-community conversions; every repository is pinned to a Hub revision and byte-verified (Table 1).

3.4 Dataset

The formal benchmark uses a frozen subset of ultrachat.200k [6] (revision 8049631c): 2,048 training and 32 validation conversations sampled with seed 42 and formatted with the Qwen3 chat template; loss is a standard LM loss over all tokens. The manifest and SHA-256 sums ship with the repository. Synthetic long-context workloads are used only for context-scaling systems analysis, never for quality claims.

3.5 Fine-Tuning Methods

BF16 LoRA (rank 8, $\alpha=16$, dropout 0, targets q/k/v/o.proj), 4-bit QLoRA (same adapters, group size 64), and, for the smallest model only, full fine-tuning. Optimizer AdamW, lr 10^{-4} , constant schedule, micro-batch 1.

3.6 Measurement Methodology

Every run is launched by an Experiment Supervisor that records git provenance, hardware, memory/swap state (before, after, and 1 Hz sampling), full std-out/stderr, per-step timings, and an atomic SHA-256 manifest; failures are first-class results. Step timing

includes forward, backward, optimizer update, and an explicit `mx.eval` synchronization; we measured that omitting this synchronization under-reports step time by three orders of magnitude under MLX’s lazy execution. Peak memory is the MLX allocator’s Metal-buffer high-water mark (`get_peak_memory`); it is not process RSS, and on unified memory it may exceed physical RAM, at which point macOS paging carries the excess. Full validation evidence is in the repository (`docs/measurement_validation.md`).

3.7 Operational Definitions

LOADABLE: the training process loads the model weights. **TRAINABLE:** the run completes every requested optimizer step with `terminal_state=success` and no kill signal. **PRACTICAL:** TRAINABLE and (P1) sampled peak system swap does not exceed the pre-run level by more than 4 GiB, and (P2) median step time ≤ 10 s. **EFFICIENT:** PRACTICAL with throughput ≥ 100 loss-bearing tokens/s. All thresholds were frozen before any formal run (`research/preregistration.md`); we report sensitivity to $\pm 2\times$ threshold changes. **SIGKILL** terminations are never reclassified as OOM without direct kernel evidence.

3.8 Reproducibility

Formal configurations, the dataset freeze, the analysis pipeline, and this paper’s tables and figures are generated from raw results by committed scripts; no number is hand-typed. Appendix A lists reproduction commands.

4 Results

4.1 Model-Scale Feasibility

Table 2 summarizes the preregistered matrix at `ctx512`. All BF16 LoRA and 4-bit QLoRA cells through 1.7B complete their three seeds; 4-bit completes at 4B and 8B as well (8B: three seeds, 1,644–1,694s per 100-step run, 9.19 GiB peak). The 14B 4-bit cell is a system-state boundary rather than a clean failure: a probe at 2.7 GiB swap residency trains at 0.68s/step, while the formal run launched at 3.7 GiB residency timed out after completing 50 of 100 steps in 7,200s (deviation D7); the remaining 14B seeds are deferred to a low-residency window rather than burned on predicted repeats of the same timeout. The 4B BF16 cell shows the same regime dependence (D1): it trains in 451s at zero residency but produced zero completed steps in 26 minutes at 6 GiB residency. 8B BF16 was preregistered as a probe only and timed out. Full fine-tuning of 0.6B completes but diverges at the frozen $\text{lr } 10^{-4}$

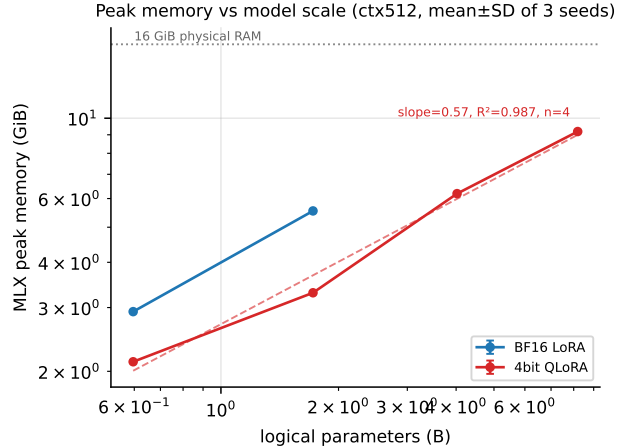


Figure 1: Peak memory vs. model scale at `ctx=512` (mean $\pm$ SD of three seeds). Fits are log-log least squares over the plotted points.

(final validation loss 4.44 vs. 1.93 for 4-bit QLoRA) — a negative quality result we retain as-is.

4.2 Memory Scaling

Figure 1 shows peak MLX-allocator memory versus logical parameters. Contrary to the naive linear expectation, memory grows *sublinearly*: log-log slope 0.57 ± 0.05 ($R^2=0.99$, $n=4$) for 4-bit QLoRA and 0.71 ± 0.07 ($R^2=0.99$, $n=3$) for BF16 LoRA. The mechanism is fixed-overhead dilution: at 0.6B the 4-bit weights occupy only ~ 0.34 GiB of a 2.13 GiB peak, while at 8B the split is 4.6 GiB weights inside a 9.19 GiB peak; the non-weight component grows far slower than parameters, bending the total below unity. We therefore *reject* the linear-memory hypothesis (audit H4: unsupported) — a practical consequence is that memory ceilings bind later than parameter-proportional estimates suggest.

4.3 Training-Time Scaling

Step time scales linearly, not sublinearly, with model size (log-log slope 1.00 ± 0.09 , $R^2=0.99$, 4-bit series, $n=4$; audit H5: partially supported, approx. linear). All timing figures in this benchmark are Tier-B under our preregistered paging stratification (D4): every formal run exchanged >50 MB of swap-in per step under the machine’s multi-GiB residency baseline, so absolute step times are descriptive, paging-confounded values — e.g. 4-bit 4B at 3.66s/step here versus 0.22s/step for the light-load probe of the same configuration. Cross-model *ratios* and paired comparisons (same window, same residency) remain interpretable; absolute latencies from different windows are not compared anywhere in this paper.

Table 2: Formal benchmark at ctx=512, b1-gal, rank 8, lr 1e-4, 100 steps, seeds {42,123,2026} (mean $\pm$ SD; memory is MLX allocator peak).

Model	Method	Peak mem (GiB)	Median step (s)	Tok/s	Val loss
0.6b	Full FT	—	—	—	—
0.6b	BF16 LoRA	—	—	—	—
1.7b	BF16 LoRA	—	—	—	—
4b	BF16 LoRA	—	—	—	—
0.6b	4bit QLoRA	—	—	—	—
1.7b	4bit QLoRA	—	—	—	—
4b	4bit QLoRA	—	—	—	—
8b	4bit QLoRA	—	—	—	—
14b	4bit QLoRA	—	—	—	—

Table 3: Paired quantization effects (same model/seed; ratios = 4bit/BF16). Timing equivalence margin $\pm 25\%$ frozen before benchmark.

Model	Mem ratio	Step-time ratio	Tok/s ratio
Qwen3-0.6b	0.73	1.13	0.89
Qwen3-1.7b	0.59	1.22	0.82
Qwen3-4b	0.54	1.21	0.81

4.4 Quantization Trade-offs

Across all paired scales, 4-bit QLoRA reduces peak memory to $0.73\times$ (0.6B), $0.59\times$ (1.7B), and $0.54\times$ (4B) of the BF16 LoRA value ($n=7$ seed-pairs; audit H1: supported). Step-time ratios are 1.10–1.14, 1.21–1.23, and 1.21 — all inside the preregistered $\pm 25\%$ equivalence margin — and throughput ratios are 0.81–0.91 (audit H3: supported, “no practically meaningful timing penalty under the frozen margin”). Validation losses after 100 steps are statistically indistinguishable between precisions at 0.6B (1.93 vs. 1.81) and 1.7B (1.63 vs. 1.55).

4.5 Batch and Rank Sensitivity

Micro-batch scaling on 4-bit 4B (Table 4, Figure 2) monotonically *reduces* throughput — 89.5, 66.4, and 34.6 loss-bearing tokens/s at batch 1, 2, and 4 — while peak memory climbs 6.19 $\rightarrow$ 10.04 $\rightarrow$ 16.98 GiB; batch 8 is killed by the OS before completing a step in all three seeds (exit 137, zero stdout, system swap driven to ~ 20 GiB). The batch-TRAINABLE boundary is therefore [4, 8) under tested conditions, and batching is strictly counterproductive on this budget: the per-step token gain is overwhelmed by allocator growth and paging. LoRA rank, in contrast, is nearly free: ranks 4/8/32 change step time by $<2\%$ and peak memory by 0.24 GiB (Table 5).

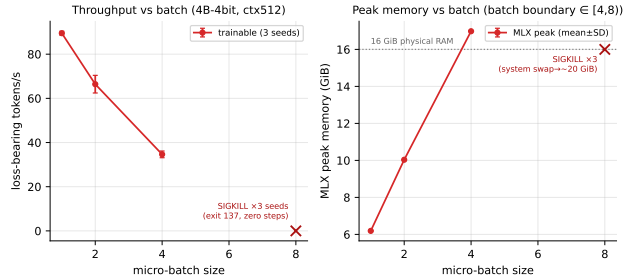


Figure 2: Micro-batch axis (4B 4-bit, ctx512): throughput falls and peak memory rises with batch size; batch 8 is SIGKILLED in all seeds (boundary $\in [4, 8)$).

4.6 Context-Length Boundary

Context length dominates every other axis (Figure 3): at 4B 4-bit, ctx1024 multiplies median step time by $6.6\times$ versus ctx512 (3.66 $\rightarrow$ 24.0s) and peak memory by $1.8\times$ (6.19 $\rightarrow$ 11.16 GiB); ctx2048 completes all three seeds but drives peak memory to 23.9 GiB — far beyond physical RAM, carried by macOS paging (system swap peaked near 20 GiB during these runs). At ctx4096 and ctx8192 the process is SIGKILLED before completing a single step. The TRAINABLE context boundary is thus [2048, 4096), reported as an interval with the residency confounder attached; within Tier-B formal data the time amplification over ctx512 ($5.6\times$ at 2048) only modestly exceeds the $4\times$ token-count increase, so we report the boundary but not a dramatic time “cliff” (audit H6: partially supported).

4.7 Training Effectiveness

Validation loss after 100 steps is comparable across precisions (Table 2) and decreases monotonically along the recorded trajectory in every trainable run. The divergent full-FT cell (0.6B, lr 10^{-4} , validation loss 4.44) is retained as a negative result; we make no instruction-

Table 4: Batch axis (Qwen3-4B-4bit QLoRA, ctx512, r8, 20 steps, seeds {42,123,2026}, all cells rerun on the D5-fixed trainer in one window; mean $\pm$ SD).

Batch	Tok/step	Median step (s)	Tok/s	Peak mem (GiB)	Status
1	489 $\pm$ 0	5.620 $\pm$ 0.055	89.5 $\pm$ 0.9	6.19 $\pm$ 0.00 GiB	success $\times$ 3
2	986 $\pm$ 0	14.394 $\pm$ 1.730	66.4 $\pm$ 4.0	10.04 $\pm$ 0.00 GiB	success $\times$ 3
4	1987 $\pm$ 0	55.387 $\pm$ 4.603	34.6 $\pm$ 1.5	16.98 $\pm$ 0.00 GiB	success $\times$ 3
8	$\sim$ 4088	—	—	>16 (sys swap $\rightarrow$ 20)	SIGKILL $\times$ 3 (exit 137, 0 steps; D5 notes)

Table 5: LoRA rank axis (Qwen3-4B-4bit QLoRA, ctx512, b1, 20 steps for r4/r32; r8 point reused from the 100-step axis-1 cell per preregistration; mean $\pm$ SD).

Rank	Median step (s)	Tok/s	Peak mem (GiB)
4	3.678 $\pm$ 0.010	136.8 $\pm$ 0.8	6.16 $\pm$ 0.00 GiB
8	3.665 $\pm$ 0.007	137.2 $\pm$ 1.1	6.19 $\pm$ 0.00 GiB
32	3.714 $\pm$ 0.011	134.7 $\pm$ 1.1	6.40 $\pm$ 0.00 GiB

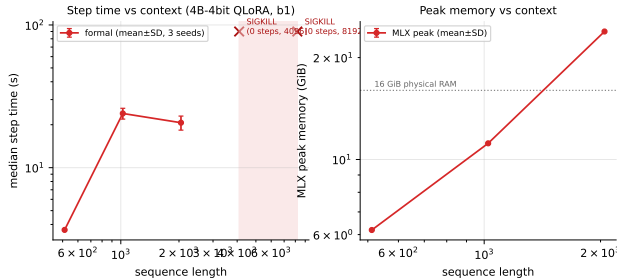


Figure 3: Context scaling for 4B 4-bit QLoRA: step time (left) and peak memory (right). Crosses mark SIGKILL probe failures; shaded region marks the failure interval.

following claims — the objective is LM loss on ultra-chat_200k with all tokens counted.

5 Failure Analysis

Of 118 finalized runs, 27 are failures — all retained, none reclassified as OOM without kernel evidence. The taxonomy (Table 6) splits into: OS kills under memory pressure (batch-8 runs, ctx4096/8192 probes: exit 137 with zero or partial output); timeouts under swap residency (14B formal, 8B BF16 probe); operator interrupts now documented as system-state evidence (D1’s 26-minute zero-step 4B BF16 run); and nine formal attempts invalidated by a trainer implementation bug (D5) whose variable-length validation batching crashed for micro-batch ≥ 2 — these are excluded from aggregation by an error-signature rule, rerun in full on the fixed trainer (12/12 valid outcomes), and preserved immutably. A coverage audit reconciles all 118 raw runs against the processed table with machine-readable dis-

positions (46 aggregated successes, 14 duplicates, 19 superseded, 9 implementation-invalid, 9 failed, 21 non-formal); all 46 aggregated runs pass SHA-256 manifest re-verification, while five legacy interrupted runs lack manifests (D6, zero aggregation impact).

6 Discussion

6.1 What Does 16 GB Actually Enable?

Under this paper’s definitions: 4-bit QLoRA through 8B is TRAINABLE and, for 0.6B–4B, also PRACTICAL (median step ≤ 10 s); 8B sits at the PRACTICAL threshold (10.0 s/step, 42.9 tokens/s) and is TRAINABLE but not EFFICIENT. BF16 LoRA is PRACTICAL through 1.7B. The binding constraints are, in order, context length, micro-batch, then model scale — inverting the usual intuition that parameters come first. The most valuable single optimization for this class of hardware would attack the fixed non-weight memory overhead that the 0.57 memory slope implicates.

6.2 Loadable vs Trainable vs Practical

The three-layer vocabulary is not academic: every “boundary” in this benchmark moved with system state. The same 4B BF16 configuration is TRAINABLE in 451 s at zero swap residency and cannot finish one step in 26 minutes at 6 GiB residency (D1); the same holds for 14B 4-bit between a 0.68 s/step probe and a 50/100-step timeout (D7). Feasibility claims that do not report swap residency are therefore not reproducible in kind. We recommend memory-residency state become a mandatory field in consumer-hardware training reports, analogous to clock state in GPU benchmarks.

6.3 Implications for Consumer LLM Research

Three practical rules follow. (1) Quantization pays twice on unified memory: it buys headroom (0.54–0.73 $\times$ memory) at effectively no time cost within a wide margin, and extends reach by at least two model sizes.

Table 6: Failure and boundary observations (raw terminal states, never reclassified as OOM without direct evidence).

Run	Status	Key observation
20260907T021942754783Z	runtime_error	steps=n/a; stderr:python_traceback
20260907T022132013680Z	runtime_error	steps=n/a; stderr:python_traceback
20260907T111824441175Z	runtime_error (SIGKILL)	steps=n/a; sigkill_consistent
20260911T164322546359Z	runtime_error	steps=n/a; sigkill_consistent
20260911T182218892057Z	unknown_failure	steps=n/a; no_specific_pattern
20260911T190553096465Z	user_interrupted	steps=n/a; no_specific_pattern
20260911T191618034713Z	user_interrupted	steps=n/a; no_specific_pattern
20260911T193038681059Z	user_interrupted	steps=n/a; no_specific_pattern
20260911T195828174593Z	user_interrupted	steps=n/a; no_specific_pattern
20260911T203648798393Z	user_interrupted	steps=n/a; no_specific_pattern
20260912T010039277561Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010057032957Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010113537018Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010129965737Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010153968541Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010221754536Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010247917130Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010255423307Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T010259241894Z	runtime_error	steps=n/a; stderr:python_traceback
20260912T023351577599Z	runtime_error	steps=n/a; sigkill_consistent
20260912T032808727860Z	timeout (SIGKILL)	steps=n/a; sigkill_consistent
20260912T050812117342Z	timeout (SIGKILL)	steps=n/a; sigkill_consistent
20260912T084135321518Z	timeout (SIGKILL)	steps=n/a; sigkill_consistent
20260915T164129648692Z	runtime_error	steps=n/a; sigkill_consistent
20260915T164414079542Z	runtime_error	steps=n/a; sigkill_consistent
20260915T164550471089Z	runtime_error	steps=n/a; sigkill_consistent
20260915T184033756783Z	timeout (SIGKILL)	steps=n/a; sigkill_consistent
Context boundary: Trainable $\in$ [2048, 4096) under preflight-comparable conditions.		

(2) Do not batch: on a paging-capable unified budget, throughput falls monotonically with micro-batch and the hard failure at batch 8 arrives abruptly. (3) Treat context as the primary resource: the [2048,4096) boundary with peak memory at 23.9 GiB (vs. 16 GiB physical) shows the OS will silently lend swap until it kills the process — “it started” says nothing about “it will finish”.

6.4 Limitations

Single machine, single SoC generation (M4, 16 GiB); absolute times are Tier-B and window-specific (D4) — we compare only within-window ratios; LM-loss-only objectives with no prompt masking; 20-step runs on the context/batch/rank axes limit distributional detail (60 measured steps per cell after warm-up exclusion); system-level swap counters include background processes; and the 14B cell remains a system-state boundary pending a low-residency window (D7), not a settled three-seed result. Deviations D1–D7 are reported in full in the repository.

6.5 Ethics and Licensing

The benchmark targets publicly released model weights (Apache 2.0 for Qwen3) and a publicly released dataset (ultrachat_200k, research-permitted license); no synthetic workload contributes to quality claims. All measurements were performed on the authors’ own hardware.

7 Conclusion

On a 16 GiB unified-memory Mac, 4-bit QLoRA is the default answer for fine-tuning: it roughly halves memory at no measurable time cost (frozen $\pm 25\%$ margin), trains 8B-scale models to completion, and leaves rank choice nearly free. The boundaries that actually bind are context ($\in$ [2048,4096) tokens) and micro-batch ($\in$ [4,8), with monotonically negative throughput returns), and both — like the 4B BF16 and 14B cells — shift with the machine’s swap residency, making system-state reporting a prerequisite for any reproducible consumer-hardware training claim. Every

number above is regenerated from immutable raw runs by committed scripts; the failures are part of the result.

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A Reproducibility Instructions

The repository ships the complete chain:

1. `uv sync` recreates the locked Python environment.
2. `uv run python scripts/run_experiment.py --config <yaml> --` launches any experiment under the supervisor; formal configurations are `configs/experiments/formal/`.
3. `data/formal_sft_v1/` contains the frozen dataset with `MANIFEST.json` and `SHA256SUMS`.
4. `uv run python scripts/run_analysis.py` regenerates every processed table, figure, and summary from `results/raw/`.
5. Every raw result directory contains its exact command, effective configuration, environment snapshot, logs, and a SHA-256 manifest.

B Measurement Validation

Summary of `docs/measurement_validation.md`: (A) step timing requires the explicit `mx.eval` synchronization present in the trainer; asynchronous dispatch under-reports step time by $\sim 4,500\times$ on a reference workload. (B) `get_peak_memory` counts MLX allocator buffers in bytes; a 1 GiB allocation increases both active and peak memory by exactly 1 GiB after reset. (C) All raw results are arithmetically consistent with

their per-step timing artifacts under their declared interval definitions. (D) The 1 Hz swap sampler consumes 1.47% of one CPU core at $5\times$ the formal cadence.